Q1. JEE Main 2026 (21 January Shift 2)

Given below are two statements :
Statement I : Crystal Field Stabilization Energy (CFSE) of $[Cr(H_2O)_6]^{2+}$ is greater than that of $[Mn(H_2O)_6]^{2+}$.
Statement II: Potassium ferricyanide has a greater spin-only magnetic moment than sodium ferrocyanide.
In the light of the above statements, choose the correct answer from the options given below :
(1) Both Statement I and Statement II are true
(2) Both Statement I and Statement II are false
(3) Statement I is true but Statement II is false
(4) Statement I is false but Statement II is true

Q2. JEE Main 2026 (21 January Shift 2)

Identify the metal ions among $Co^{2+}, Ni^{2+}, Fe^{2+}, V^{3+}$ and Ti^{2+} having a spin-only magnetic moment value more than 3.0 BM. The sum of unpaired electrons present in the high spin octahedral complexes formed by those metal ions is ____.

Q3. JEE Main 2026 (23 January Shift 1)

Given below are two statements :
Statement I: $[CoBr_4]^{2-}$ ion will absorb light of lower energy than $[CoCl_4]^{2-}$ ion.
Statement II : In $[CoI_4]^{2-}$ ion, the energy separation between the two set of d-orbitals is more than $[CoCl_4]^{2-}$ ion.
In the light of the above statements, choose the correct answer from the options given below :
(1) Both Statement I and Statement II are false
(2) Both Statement I and Statement II are true
(3) Statement I is false but Statement II is true
(4) Statement I is true but Statement II is false

Q4. JEE Main 2026 (23 January Shift 1)

The statements that are incorrect about the nickel(II) complex of dimethylglyoxime are :
A. It is red in colour.
B. It has a high solubility in water at $pH = 9$.
C. The Ni ion has two unpaired d-electrons.
D. The $N - Ni - N$ bond angle is almost close to 90°.
E. The complex contains four five-membered metallacycles (metal containing rings).
Choose the correct answer from the options given below :
(1) C and D Only
(2) C and E Only
(3) B, C and E Only
(4) A, D and B Only

Q5. JEE Main 2026 (23 January Shift 1)
The crystal field splitting energy of $[\mathrm{Co}(\text{ oxalate })_3]^{3-}$ complex is ' n′ times that of the $[\mathrm{Cr}(\text{ oxalate })_3]^{3-}$ complex. Here ' n′ is ____. (Assume $\Delta_0 \gg \mathrm{P}$)

Q6. JEE Main 2026 (28 January Shift 2)
The correct increasing order of spin-only magnetic moment values of the complex ions $[\mathrm{MnBr}_4]^{2-}(\mathrm{A}), [\mathrm{Cu}(\mathrm{H_2O})_6]^{2+}(\mathrm{B}), [\mathrm{Ni}(\mathrm{CN})_4]^{2-}(\mathrm{C})$ and $[\mathrm{Ni}(\mathrm{H_2O})_6]^{2+}(\mathrm{D})$ is :
(1) $\mathrm{C} = \mathrm{D} < \mathrm{B} < \mathrm{A}$
(2) $\mathrm{C} < \mathrm{B} < \mathrm{D} < \mathrm{A}$
(3) $\mathrm{A} = \mathrm{B} < \mathrm{D} < \mathrm{C}$
(4) $A = B < C < D$

Q7. JEE Main 2026 (28 January Shift 2)
The number of isoelectronic species among $\mathrm{Sc}^{3+}, \mathrm{Cr}^{2+}, \mathrm{Mn}^{3+}, \mathrm{Co}^{3+}$ and Fe^{3+} is ' n′. If ' n′ moles of AgCl is formed during the reaction of complex with formula $\mathrm{CoCl_3(en)_2NH_3}$ with excess of $\mathrm{AgNO_3}$ solution, then the number of electrons present in the t_{2g} orbital of the complex is ____.

Q8. JEE Main 2026 (24 January Shift 2)
The wavelength of light absorbed for the following complexes are in the order
$\underset{(\mathrm{I})}{[\mathrm{Co}(\mathrm{NH_3})_6]^{3+}}; \underset{(\mathrm{II})}{[\mathrm{Co}(\mathrm{H_2O})_6]^{3+}}; \underset{(\mathrm{III})}{[\mathrm{Co}(\mathrm{CN})_6]^{3-}}; \underset{(\mathrm{IV})}{[\mathrm{Co}(\mathrm{NH_3})_5(\mathrm{H_2O})]^{3+}}; \underset{(\mathrm{V})}{[\mathrm{CoF_6}]^{3-}}$
(1) $\mathrm{III} < \mathrm{I} < \mathrm{IV} < \mathrm{V} < \mathrm{II}$
(2) $\mathrm{III} < \mathrm{I} < \mathrm{IV} < \mathrm{II} < \mathrm{V}$
(3) $\mathrm{III} < \mathrm{I} < \mathrm{II} < \mathrm{IV} < \mathrm{V}$
(4) $\mathrm{III} < \mathrm{IV} < \mathrm{I} < \mathrm{II} < \mathrm{V}$

Q9. JEE Main 2026 (24 January Shift 2)
A chromium complex with a formula $\mathrm{CrCl_3 \cdot 6H_2O}$ has a spin only magnetic moment value of 3.87 BM and its solution conductivity corresponds to 1 : 2 electrolyte. 2.75 g of the complex solution was initially passed through a cation exchanger. The solution obtained after the process was reacted with excess of $\mathrm{AgNO_3}$. The amount of AgCl formed in the above process is ____ g. (Nearest integer)
[Given: Molar mass in $\mathrm{gmol}^{-1}\mathrm{Cr} : 52; \mathrm{Cl} : 35.5, \mathrm{Ag} : 108, \mathrm{O} : 16, \mathrm{H} : 1$]

Q10. JEE Main 2026 (22 January Shift 2)

$[Ni(PPh_3)_2Cl_2]$ is a paramagnetic complex. Identify the INCORRECT statements about this complex.

A. The complex exhibits geometrical isomerism.

B. The complex is white in colour.

C. The calculated spin-only magnetic moment of the complex is 2.84 BM .

D. The calculated CFSE (Crystal Field Stabilization Energy) of Ni in this complex is $-0.8\Delta_o$.

E. The geometrical arrangement of ligands in this complex is similar to that in $Ni(CO)_4$.

Choose the correct answer from the options given below :

(1) C, D and E Only
(2) C and D Only
(3) A, B and D Only
(4) A and B Only

Q11. JEE Main 2026 (22 January Shift 1)

Consider the transition metal ions Mn^{3+}, Cr^{3+}, Fe^{3+} and Co^{3+} and all form low spin octahedral complexes. The correct decreasing order of unpaired electrons in their respective d-orbitals of the complexes is

(1) $Mn^{3+} > Fe^{3+} > Co^{3+} > Cr^{3+}$
(2) $Cr^{3+} > Mn^{3+} > Fe^{3+} > Co^{3+}$
(3) $Fe^{3+} > Co^{3+} > Mn^{3+} > Cr^{3+}$
(4) $Cr^{3+} > Fe^{3+} > Co^{3+} > Mn^{3+}$

Q12. JEE Main 2026 (23 January Shift 2)

Identify the CORRECT set of details from the following:

A. $[Co(NH_3)_6]^{3+}$: Inner orbital complex; d^2sp^3 hybridized

B. $[MnCl_6]^{3-}$: Outer orbital complex; $sp^3 d^2$ hybridized

C. $[CoF_6]^{3-}$: Outer orbital complex; d^2sp^3 hybridized

D. $[FeF_6]^{3-}$: Outer orbital complex; $sp^3 d^2$ hybridized

E. $[Ni(CN)_4]^{2-}$: Inner orbital complex; sp^3 hybridized

Choose the correct answer from the options given below:

(1) A, C & E Only
(2) A, B & D Only
(3) C & D Only
(4) A, B, C, D & E

Q13. JEE Main 2026 (23 January Shift 2)

Total number of unpaired electrons present in the central metal atoms/ions of $[Ni(CO)_4]$, $[NiCl_4]^{2-}$, $[PtCl_2(NH_3)_2]$, $[Ni(CN)_4]^{2-}$ and $[Pt(CN)_4]^{2-}$ is ____.

Q14. JEE Main 2026 (21 January Shift 1)

Given below are two statements :

Statement I : Among $[Cu(NH_3)_4]^{2+}$, $[Ni(en)_3)]^{2+}$, $[Ni(NH_3)_6]^{2+}$ and $[Mn(H_2O)_6]^{2+}$, $[Mn(H_2O)_6]^{2+}$ has the maximum number of unpaired electrons.

Statement II : The number of pairs among $\left\{[NiCl_4]^{2-}, [Ni(CO)_4]\right\}$, $\left\{[NiCl_4]^{2-}, [Ni(CN)_4]^{2-}\right\}$ and $\left\{[Ni(CO)_4], [Ni(CN)_4]^{2-}\right\}$ that contain only diamagnetic species is two.

In the light of the above statements, choose the correct answer from the options given below : Statement I is true but Statement II is false Both Statement I and Statement II are false

(1) Statement I is true but Statement II is false

(2) Both Statement I and Statement II are false

(3) Both Statement I and Statement II are true

(4) Statement I is false but Statement II is true

Q15. JEE Main 2026 (28 January Shift 1)

The correct statement among the following is:

(1) $Ni(CO)_4$ and $[Ni(CN)_4]^{2-}$ are diamagnetic and $[NiCl_4]^{2-}$ is paramagnetic.

(2) $Ni(CO)_4$ and $[NiCl_4]^{2-}$ are diamagnetic and $[Ni(CN)_4]^{2-}$ is paramagnetic.

(3) $[Ni(CN)_4]^{2-}$ and $[NiCl_4]^{2-}$ are diamagnetic and $Ni(CO)_4$ is paramagnetic.

(4) $Ni(CO)_4$ is diamagnetic and $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ are paramagnetic.

Q16. JEE Main 2026 (28 January Shift 1)

X is the number of geometrical isomers exhibited by $[Pt(NH_3)(H_2O)BrCl]$.

Y is the number of optically inactive isomer(s) exhibited by $[CrCl_2(ox)_2]^{3-}$

Z is the number of geometrical isomers exhibited by $[Co(NH_3)_3(NO_2)_3]$.

The value of $X + Y + Z$ is ____.

Q17. JEE Main 2026 (24 January Shift 1)

Given below are two statements:

Statement I: Hybridisation, shape and spin only magnetic moment of $K_3[Co(CO_3)_3]$ is sp^3d^2, octahedral and 4.9 BM respectively.

Statement II: Geometry, hybridisation and spin only magnetic moment values (BM) of the ions $[Ni(CN)_4]^{2-}$, $[MnBr_4]^{2-}$ and $[CoF_6]^{3-}$ respectively are square planar, tetrahedral, octahedral; dsp^2, sp^3, sp^3d^2 and $0, 5.9, 4.9$.

In the light of the above statements, choose the correct answer from the options given below

(1) Statement I is true but Statement II is false

(2) Both Statement I and Statement II are false

(3) Both Statement I and Statement II are true

(4) Statement I is false but Statement II is true

Q18. JEE Main 2026 (24 January Shift 1)

Given below are two statements:

Statement I: The number of paramagnetic species among $[CoF_6]^{3-}$, $[TiF_6]^{3-}$, V_2O_5 and $[Fe(CN)_6]^{3-}$ is 3.

Statement II: $K_4[Fe(CN)_6] < K_3[Fe(CN)_6] < [Fe(H_2O)_6]SO_4 \cdot H_2O < [Fe(H_2O)_6]Cl_3$ is the correct order in terms of number of unpaired electron(s) present in the complexes.

In the light of the above statements, choose the correct answer from the options given below

(1) Both Statement I and Statement II are true

(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are false

(4) Statement I is true but Statement II is false

Q19. JEE Main 2026 (24 January Shift 1)

Consider a mixture ' X ' which is made by dissolving 0.4 mol of $[Co(NH_3)_5SO_4]Br$ and 0.4 mol of $[Co(NH_3)_5Br]SO_4$ in water to make 4 L of solution. When 2 L of mixture ' X ' is allowed to react with excess of $AgNO_3$, it forms precipitate ' Y '. The rest 2 L of mixture ' X ' reacts with excess $BaCl_2$ to form precipitate ' Z '. Which of the following statements is CORRECT?

(1) 0.1 mol of ' Y ' is formed.

(2) 0.2 mol of ' Z ' is formed.

(3) 0.4 mol of ' Z ' is formed.

(4) 'Y' is $BaSO_4$ and ' Z ' is AgBr.

ANSWERS AND SOLUTIONS

1. (1)	**2.** 7	**3.** (4)	**4.** (3)	**5.** 2	**6.** (2)	**7.** 6	**8.** (2)
9. 3	**10.** (3)	**11.** (2)	**12.** (2)	**13.** 2	**14.** (1)	**15.** (1)	**16.** 6
17. (3)	**18.** (1)	**19.** (2)					

1. **(1)** $\left[\mathrm{Mn(H_2O)_6}\right]^{2+} \Rightarrow$ CFSE value is zero because of d^5 configuration with Weak Field Ligands in coordination number 6 $\left[\mathrm{Cr(H_2O)_6}\right]^{2+}$.

$\left[\mathrm{Cr(H_2O)_6}\right]^{2+} \Rightarrow$ CFSE value is $-0.6\Delta_0$ because of d^4 configuration with Weak Field Ligands in coordination number 6.

For : $\mathrm{K_3\,[Fe(CN)_6]}$, $\mu = \sqrt{1(1+2)} = \sqrt{3}$ B.M.

For : $\mathrm{Na_4\,[Fe(CN)_6]}$, $\mu = \sqrt{0}$ B.M.

2. **(7)** The spin-only magnetic moment is given by $\mu = \sqrt{n(n+2)}$ BM, where n is the number of unpaired electrons.
For $\mu > 3.0$ BM, n must be ≥ 3 (since for $n = 2$, $\mu = \sqrt{8} \approx 2.83$ BM and for $n = 3$, $\mu = \sqrt{15} \approx 3.87$ BM).
Electronic configurations and unpaired electrons (n) in high spin octahedral complexes:

Metal Ion	Configuration	n	μ (BM)
Co^{2+}	$[Ar]3d^7$	3	3.87
Ni^{2+}	$[Ar]3d^8$	2	2.83
Fe^{2+}	$[Ar]3d^6$	4	4.90
V^{3+}	$[Ar]3d^2$	2	2.83
Ti^{2+}	$[Ar]3d^2$	2	2.83

The metal ions with $\mu > 3.0$ BM are Co^{2+} ($n = 3$) and Fe^{2+} ($n = 4$).
The sum of unpaired electrons for these ions is $3 + 4 = 7$.

3. **(4)** Crystal field theory explains d-d transitions in complexes based on ligand field strength.
Statement I: Ligand field strength order is $I^- < Br^- < Cl^- < F^-$. Since Br^- is weaker than Cl^-, $[CoBr_4]^{2-}$ has smaller crystal field splitting (Δ) than $[CoCl_4]^{2-}$. The d-d absorption energy equals the splitting energy, so $[CoBr_4]^{2-}$ absorbs lower energy light. TRUE
Statement II: I^- is a much weaker field ligand than Cl^-. Therefore, $[CoI_4]^{2-}$ has smaller crystal field splitting than $[CoCl_4]^{2-}$, not larger. The energy separation in $[CoI_4]^{2-}$ is less than in $[CoCl_4]^{2-}$. FALSE
Statement I is true but Statement II is false.

4. **(3)** Statement A is incorrect as the complex is yellow-green, not red.
Statement B is incorrect because the complex shows low solubility in water, especially at neutral pH.
Statement C is incorrect because Ni(II) in square planar geometry has all d-electrons paired (d^8 configuration), not unpaired electrons.
Statement D is correct regarding the N-Ni-N bond angle being approximately 90° in square planar geometry.
Statement E is correct as the complex contains two five-membered metallacycles (chelate rings) per dimethylglyoxime molecule.
The incorrect statements are B, C, and E.

5. **(2)** The Crystal Field Stabilization Energy (CFSE) is calculated based on the electronic configuration of the central metal ion in the octahedral field.
For $[Co(ox)_3]^{3-}$:
Cobalt is in $+3$ oxidation state. Co^{3+} has a d^6 configuration.
Given $\Delta_0 \gg P$, it is a low spin complex. The electrons fill the t_{2g} level first.
Configuration: $t_{2g}^6 e_g^0$.
CFSE for $Co^{3+} = 6 \times (-0.4\Delta_0) + 0 \times (0.6\Delta_0) = -2.4\Delta_0$.
For $[Cr(ox)_3]^{3-}$:
Chromium is in $+3$ oxidation state. Cr^{3+} has a d^3 configuration.
Configuration: $t_{2g}^3 e_g^0$.
CFSE for $Cr^{3+} = 3 \times (-0.4\Delta_0) + 0 \times (0.6\Delta_0) = -1.2\Delta_0$.
Comparing the magnitudes of the stabilization energies:
Magnitude of CFSE for $Co^{3+} = 2.4\Delta_0$.
Magnitude of CFSE for $Cr^{3+} = 1.2\Delta_0$.
The ratio $n = \frac{2.4\Delta_0}{1.2\Delta_0} = 2$.

6. **(2)** Calculate spin-only magnetic moment using $\mu_s = \sqrt{n(n+2)}$ where n is the number of unpaired electrons.
For $[MnBr_4]^{2-}$ (A): Mn(II) in tetrahedral environment is high spin with 5 unpaired electrons.
$\mu_s = \sqrt{5(7)} = \sqrt{35} \approx 5.92$ BM.
For $[Cu(H_2O)_6]^{2+}$ (B): Cu(II) has 1 unpaired electron. $\mu_s = \sqrt{1(3)} = \sqrt{3} \approx 1.73$ BM.
For $[Ni(CN)_4]^{2-}$ (C): Ni(II) in square planar geometry is low spin with 0 unpaired electrons. $\mu_s = 0$ BM.
For $[Ni(H_2O)_6]^{2+}$ (D): Ni(II) in octahedral environment is high spin with 2 unpaired electrons.
$\mu_s = \sqrt{2(4)} = \sqrt{8} \approx 2.83$ BM.
Increasing order: C (0) < B (1.73) < D (2.83) < A (5.92)

7. **(6)** For the complex $CoCl_3(en)_2NH_3$, cobalt is in +3 oxidation state with electron configuration $[Ar]3d^6$.
In an octahedral field with strong field ligands (ethylenediamine, ammonia, chloride), Co^{3+} forms a low-spin complex with $t_{2g}^6 e_g^0$ configuration.
The t_{2g} orbitals consist of d_{xy}, d_{xz}, and d_{yz}.
With all 6 electrons filling the three t_{2g} orbitals, the total number of electrons in the t_{2g} orbital set is 6.

8. **(2)** Wavelength of light absorbed increases as C.F.S.E of complex decreases.
$[Co(CN)_6]^{3-}$ has maximum CFSE
$[CoF_6]^{3-}$ has least CFSE
Ligand field strength ↑ ; C.F.S.E ↑
Correct wavelength order.
$V > II > IV > I > III$

9. **(3)** The complex $CrCl_3 \cdot 6H_2O$ with magnetic moment 3.87 BM indicates $3.87 = \sqrt{n(n+2)}$ giving n = 3 unpaired electrons, confirming Cr^{3+}.
A 1:2 electrolyte means structure is $[Cr(H_2O)_5Cl]^{2+}Cl_2^-$.
Molar mass = 52 + 5(18) + 35.5 + 6(18) = 285.5 g/mol.
Moles of complex = 2.75/285.5 = 0.00963 mol.
After cation exchange, 2 Cl^- remain per formula unit (from the 1:2 electrolyte), giving 0.00963 × 2 = 0.01926 mol of free Cl^-.
Reacting with excess $AgNO_3$: mass of AgCl = 0.01926 × 143.5 = 2.76 g ≈ 3 g (nearest integer).

10. **(3)** In $[Ni(PPh_3)_2Cl_2]$, Nickel is in $+2$ oxidation state. Ni^{2+} has a $3d^8$ electronic configuration.
The complex is given as paramagnetic. For a 4-coordinate d^8 system, paramagnetism implies a tetrahedral geometry (sp^3 hybridization) with two unpaired electrons.
Statement A: Tetrahedral complexes of the type $[MA_2B_2]$ do not exhibit geometrical isomerism because all positions are adjacent to each other. Thus, A is incorrect.
Statement B: Most $Ni(II)$ complexes are colored due to $d-d$ transitions. $[Ni(PPh_3)_2Cl_2]$ is a dark blue/green crystalline solid, not white. Thus, B is incorrect.
Statement C: With 2 unpaired electrons, the spin-only magnetic moment
$\mu = \sqrt{n(n+2)} = \sqrt{2(2+2)} = \sqrt{8} \approx 2.83$ BM. Thus, C is correct.
Statement D: For a tetrahedral complex with d^8 configuration, the CFSE is calculated as
$CFSE = [(-0.6 \times 4) + (0.4 \times 4)]\Delta_t = -0.8\Delta_t$. The statement mentions $-0.8\Delta_o$, which is incorrect as the complex is tetrahedral, not octahedral. Thus, D is incorrect.
Statement E: $Ni(CO)_4$ is a tetrahedral complex (sp^3 hybridization). Since $[Ni(PPh_3)_2Cl_2]$ is also tetrahedral, their geometrical arrangements are similar. Thus, E is correct.
The incorrect statements are A, B, and D.

11. **(2)** For low spin octahedral complexes, electrons fill the t_{2g} orbitals first before pairing or moving to e_g orbitals. The configuration depends on the number of d-electrons.
For Cr^{3+}: Electronic configuration is $3d^3$. In an octahedral field, the configuration is $t_{2g}^3e_g^0$. Number of unpaired electrons (n) = 3.
For Mn^{3+}: Electronic configuration is $3d^4$. For low spin, the configuration is $t_{2g}^4e_g^0$. Number of unpaired electrons (n) = 2.
For Fe^{3+}: Electronic configuration is $3d^5$. For low spin, the configuration is $t_{2g}^5e_g^0$. Number of unpaired electrons (

$n) = 1$.

For Co^{3+}: Electronic configuration is $3d^6$. For low spin, the configuration is $t_{2g}^6 e_g^0$. Number of unpaired electrons ($n) = 0$.

Comparing the number of unpaired electrons:

$Cr^{3+}(3) > Mn^{3+}(2) > Fe^{3+}(1) > Co^{3+}(0)$.

The correct decreasing order is $Cr^{3+} > Mn^{3+} > Fe^{3+} > Co^{3+}$.

12. (2) Analyzing each complex based on Valence Bond Theory (VBT):

A. $[Co(NH_3)_6]^{3+}$:

Co^{3+} is $3d^6$. NH_3 is a strong field ligand (SFL) for Co^{3+}, causing pairing. This leaves two $3d$ orbitals vacant. Hybridization is d^2sp^3 (Inner orbital complex). Statement A is correct.

B. $[MnCl_6]^{3-}$:

Mn^{3+} is $3d^4$. Cl^- is a weak field ligand (WFL). No pairing occurs. To accommodate 6 ligands, it uses $4s$, $4p$, and $4d$ orbitals. Hybridization is sp^3d^2 (Outer orbital complex). Statement B is correct.

C. $[CoF_6]^{3-}$:

Co^{3+} is $3d^6$. F^- is a WFL. No pairing occurs. Hybridization is sp^3d^2 (Outer orbital complex). Statement C is incorrect because it mentions d^2sp^3.

D. $[FeF_6]^{3-}$:

Fe^{3+} is $3d^5$. F^- is a WFL. No pairing occurs. Hybridization is sp^3d^2 (Outer orbital complex). Statement D is correct.

E. $[Ni(CN)_4]^{2-}$:

Ni^{2+} is $3d^8$. CN^- is a SFL, causing pairing of the two unpaired electrons in $3d$. This leaves one $3d$ orbital vacant. Hybridization is dsp^2 (Square planar, Inner orbital complex). Statement E is incorrect because it mentions sp^3.

Correct statements are A, B, and D.

13. (2) $[Ni(CO)_4]$:

Ni is zero-valent with configuration $[Ar]3d^{10}4s^2$.

CO is a strong field ligand causing complete pairing in the filled $3d^{10}$ orbital.

Unpaired electrons = 0.

$[NiCl_4]^{2-}$:

Ni is $+2$ with configuration $[Ar]3d^8$.

Cl is a weak field ligand producing high-spin tetrahedral geometry with configuration $3d^8$: five orbitals with two

electrons paired, three electrons unpaired.
Unpaired electrons = 2.

$[PtCl_2(NH_3)_2]$:
Pt is $+2$ with configuration $[Xe]4f^{14}5d^8$.
Pt(II) adopts square planar geometry with strong-field splitting leading to complete pairing: $5d^8$ configuration places all electrons in bonding orbitals.
Unpaired electrons = 0.

$[Ni(CN)_4]^{2-}$:
Ni is $+2$ with $3d^8$ configuration.
CN is a strong field ligand.
Square planar geometry is preferred, resulting in all electrons paired.
Unpaired electrons = 0.

$[Pt(CN)_4]^{2-}$:
Pt is $+2$ with $5d^8$ configuration.
CN is a strong field ligand producing square planar geometry with complete pairing.
Unpaired electrons = 0.

Total unpaired electrons = 0 + 2 + 0 + 0 + 0 = 2.

14. (1) Statement I: Unpaired electrons in each complex:
$[Cu(NH_3)_4]^{2+}$: Cu^{2+} is $d^9 \rightarrow$ 1 unpaired
$[Ni(en)_3]^{2+}$: Ni^{2+} is d^8, octahedral $\rightarrow$ 2 unpaired
$[Ni(NH_3)_6]^{2+}$: Ni^{2+} is $d^8 \rightarrow$ 2 unpaired
$[Mn(H_2O)_6]^{2+}$: Mn^{2+} is d^5, weak field $\rightarrow$ high spin $\rightarrow$ 5 unpaired
$[Mn(H_2O)_6]^{2+}$ has maximum. Statement I is true.
Statement II: $[NiCl_4]^{2-}$: Ni^{2+} (d^8), tetrahedral $\rightarrow$ paramagnetic
$[Ni(CO)_4]$: Ni^0 (d^{10}) $\rightarrow$ diamagnetic
$[Ni(CN)_4]^{2-}$: Ni^{2+} (d^8), square planar $\rightarrow$ diamagnetic
Only pair $\{[Ni(CO)_4], [Ni(CN)_4]^{2-}\}$ has both diamagnetic. Count = 1, not 2. Statement II is false.

15. (1) In $Ni(CO)_4$, Nickel is in 0 oxidation state with configuration $[Ar]3d^8 4s^2$. CO is a strong field ligand, causing pairing of $4s$ electrons into $3d$ orbitals, resulting in $3d^{10}$ configuration. It is sp^3 hybridized and diamagnetic (zero unpaired electrons).
In $[Ni(CN)_4]^{2-}$, Nickel is in $+2$ oxidation state with configuration $[Ar]3d^8$. CN^- is a strong field ligand, causing pairing of the two unpaired electrons in $3d$ orbitals. It is dsp^2 hybridized (square planar) and diamagnetic.
In $[NiCl_4]^{2-}$, Nickel is in $+2$ oxidation state with configuration $[Ar]3d^8$. Cl^- is a weak field ligand and cannot

cause pairing of electrons. It has two unpaired electrons in $3d$ orbitals, making it sp^3 hybridized (tetrahedral) and paramagnetic.

Therefore, $Ni(CO)_4$ and $[Ni(CN)_4]^{2-}$ are diamagnetic, while $[NiCl_4]^{2-}$ is paramagnetic.

16. (6) For $[Pt(NH_3)(H_2O)BrCl]$ with square planar geometry and four different ligands, geometrical isomerism arises from different positional arrangements: X = 3.

For $[CrCl_2(ox)_2]^{3-}$, the trans isomer is optically inactive (meso compound) while cis is optically active, giving Y = 1.

For $[Co(NH_3)_3(NO_2)_3]$ in octahedral geometry, facial (fac) and meridional (mer) isomers are possible: Z = 2.

Therefore, X + Y + Z = 3 + 1 + 2 = 6

17. (3) In Statement I, for $K_3[Co(CO_3)_3]$, Cobalt is in $+3$ oxidation state (Co^{3+} : $3d^6$). Carbonate (CO_3^{2-}) is a weak field ligand. Thus, no pairing occurs, resulting in 4 unpaired electrons. Hybridisation is sp^3d^2, shape is octahedral, and $\mu = \sqrt{4(4+2)} = \sqrt{24} \approx 4.9$ BM. Statement I is true.

In Statement II:

For $[Ni(CN)_4]^{2-}$, Ni^{2+} is $3d^8$. CN^- is a strong field ligand, causing pairing. Hybridisation is dsp^2, geometry is square planar, and $\mu = 0$ BM.

For $[MnBr_4]^{2-}$, Mn^{2+} is $3d^5$. Br^- is a weak field ligand. Hybridisation is sp^3, geometry is tetrahedral, and $\mu = \sqrt{5(5+2)} = \sqrt{35} \approx 5.9$ BM.

For $[CoF_6]^{3-}$, Co^{3+} is $3d^6$. F^- is a weak field ligand. Hybridisation is sp^3d^2, geometry is octahedral, and $\mu = \sqrt{4(4+2)} = \sqrt{24} \approx 4.9$ BM.

Statement II is true.

Both statements are true.

18. (1) Statement I: Analysis of paramagnetic species.

1. $[CoF_6]^{3-}$: Co^{3+} is $3d^6$. F^- is a weak field ligand (WFL), so configuration is $t_{2g}^4 e_g^2$. Unpaired electrons (n) = 4. Paramagnetic.
2. $[TiF_6]^{3-}$: Ti^{3+} is $3d^1$. $n = 1$. Paramagnetic.
3. V_2O_5: V^{5+} is $3d^0$. $n = 0$. Diamagnetic.
4. $[Fe(CN)_6]^{3-}$: Fe^{3+} is $3d^5$. CN^- is a strong field ligand (SFL), so configuration is $t_{2g}^5 e_g^0$. $n = 1$. Paramagnetic.

Total paramagnetic species = 3. Statement I is true.

Statement II: Analysis of unpaired electrons.

1. $K_4[Fe(CN)_6]$: Fe^{2+} ($3d^6$) with SFL $\rightarrow t_{2g}^6 e_g^0$, $n = 0$.
2. $K_3[Fe(CN)_6]$: Fe^{3+} ($3d^5$) with SFL $\rightarrow t_{2g}^5 e_g^0$, $n = 1$.
3. $[Fe(H_2O)_6]SO_4 \cdot H_2O$: Fe^{2+} ($3d^6$) with WFL $\rightarrow t_{2g}^4 e_g^2$, $n = 4$.
4. $[Fe(H_2O)_6]Cl_3$: Fe^{3+} ($3d^5$) with WFL $\rightarrow t_{2g}^3 e_g^2$, $n = 5$.

Order of n: $0 < 1 < 4 < 5$. Statement II is true.

Both statements are true.

19. (2) 0.4 mol $[Co(NH_3)_5SO_4]Br$ + 0.4 mol $[Co(NH_3)_5Br]SO_4$ are present in 4 lit. solution.

2 lit. of mixture will contain 0.2 mol of each complex.

2 lit. mixture on reaction with excess $AgNO_3$, 0.2 mole $AgBr$ will be formed (Y).

2 lit. mixture on reaction with excess $BaCl_2$, 0.2 mole $BaSO_4$ will be formed (Z).